

Nikhil Vidhani

Lead Data Scientist – Digital Fulfilment & Simulations, Target Corporation

Executive Summary

- Architected Nemo, an end-to-end simulation platform for order allocation and available-to-promise, now the primary environment for scenario testing and policy design across Target's digital fulfilment network, potentially impacting up to \$8B in digital sales.
- Built and deployed Nemo AI, a production multi-agent assistant for simulation research and diagnosis, while driving agentic AI adoption and sound AI engineering practices across the data-science organisation.
- Conceived and led the revamp of Target's legacy consolidation algorithm, now in A/B-test preparation, with projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental sales.
- Deep technical expertise in Python and R, distributed data processing (Spark, Kafka, Hive/Hadoop, and SQL), algorithm design, machine learning, and software engineering, supported by a Ph.D. in Finance from IIM Bangalore and a master's degree in engineering from IISc Bangalore.
- Cross-functional and people leader partnering with product, engineering, and operations, mentoring a five-person data-science team and previously leading a finance and audit analytics team at WNS.

Industry and Research Experience

- Nov 2022 – **Lead Data Scientist, Target Corporation, Bangalore, Team Size: 5.**
- Present
- Own the strategy, design, and expansion of end-to-end simulation capabilities for order allocation and available-to-promise, enabling network-wide evaluation of inventory, capacity, fulfilment, and promise policies.
 - Develop cost-versus-speed optimisation models and lead large-scale simulation studies to inform consolidation, promise-speed, and fulfilment decisions with multi-million-dollar potential impact on shipping costs and digital sales.
 - Lead product and technical roadmaps for simulation platforms and algorithms, partnering with engineering on architecture, productionisation, validation, release planning, and operational support.
 - Serve as *AI Champion* for the Digital Fulfilment data-science organisation—deploying a production generative AI assistant, building an internal MCP service and reusable AI capabilities, and promoting sound practices for agentic AI development, evaluation, observability, and responsible adoption.
 - Partner with product, engineering, supply-chain, and fulfilment stakeholders to translate ambiguous business problems into experiments and production-ready solutions, while mentoring data scientists on technical design, experimentation, and stakeholder communication.
- Feb 2022 – **Sr. Group Manager (Data Science), WNS Global Services, Bangalore, Team Size: 5.**
- Nov 2022
- Led a five-member data-science team delivering analytics and machine-learning solutions for finance and accounting use cases (payments, collections, reporting, prediction, and auditing) for global clients, driving multi-million-dollar operational savings.
 - Owned the end-to-end lifecycle, from problem framing with business stakeholders through data engineering, modelling, API development, and automated deployment, ensuring that solutions were reliable and maintainable.
 - Maintained shared GitHub repositories, code-quality standards, and CI/CD workflows; mentored junior team members through code reviews, technical sessions, and structured knowledge-sharing.

- Jun 2016 – **Research Scholar**, *Indian Institute of Management, Bangalore*, IC.
- Jan 2022
- Researched asset-pricing anomalies and examined the impact of disagreement on trading volume, analyst forecasts, and 10-K document characteristics.
 - Conducted comprehensive data collection, cleaning, modelling, and statistical analysis.
 - Published multiple papers and presented findings at several conferences. *Complete list available at: publications_CV.pdf*
- Jun 2018 – **Primary Instructor**, *Programming and Data Analysis*, Batch Size: 5–40.
- Jan 2022
- Designed and taught multiple short- and long-format courses in R programming, data analysis, and LaTeX.
 - Instructor rating: 4.7/5 (<https://github.com/nik141088/applied-R>).
- Jul 2012 – **Software Engineer**, *Cisco Systems, Bangalore*, IC.
- May 2016
- Designed, implemented, tested, reviewed, and documented software for 4G-LTE and Wi-Fi systems.
 - Contributed to high-availability, redundancy, and network-security architecture.

Major Projects

- 2026 – **Nemo AI & Generative AI Enablement**, *Target*.
- Present
- Conceived, built, and deployed **Nemo AI**, an internally hosted multi-agent orchestrator/executor/synthesiser that performs deep research on simulation runs, answers conceptual questions, and analyses simulation inputs and outputs to accelerate debugging.
 - Engineered production-grade capabilities, including multi-environment authentication, per-request cost and latency tracking, persisted chat history, and full interaction traces for observability, governance, and evaluation-driven improvement.
 - Prototyped a specification-driven “LLM Wiki” knowledge-curation approach (an alternative to RAG) for medium-sized internal knowledge bases, with immutable source citations to improve traceability and reduce hallucinations.
 - Built a Model Context Protocol (MCP) service integrating multiple internal systems for simulation-run diagnosis and data analysis; developed reusable skills and agents for root-cause analysis, code review, and PR generation, dramatically reducing manual analysis effort.
 - Led weekly AI adoption forums and share-outs, promoting specification-driven development, testing, prompt versioning, golden-set evaluation design, observability, and human ownership of AI-generated code.
- 2025 – **Consolidation Algorithm Revamp**, *Target*.
- Present
- Redesigned Target’s consolidation algorithm to jointly optimise customer promise speed and shipping cost across the last-mile network, replacing legacy heuristics with a principled modelling framework.
 - Leveraged Nemo’s integrated Promise and GOA simulations to evaluate policy alternatives under realistic order, inventory, and capacity patterns through scenario and sensitivity analysis.
 - Led functional validation across data-science and engineering implementations, increasing agreement from 20% to 95% and identifying defects in both; progressed to mirror testing and designed a controlled A/B test across 10 ZIP codes, with speed, cost, and No-Rush offer rate as core metrics.
 - Quantified projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental digital sales through improved conversion and margin.
 - Implemented, tested, and tuned the algorithm for a latency-critical promise service, meeting strict performance and reliability constraints.
 - Secured alignment across product, engineering, and fulfilment through decision documents, technical presentations, and knowledge-transfer sessions, enabling phased rollout planning.

2023 – Present	Nemo: Unified Simulation Platform, Target. ○ Led a five-member team to design, build, and operate Nemo, a unified simulation platform for last-mile promise and fulfilment, supporting more than 100 stakeholders across planning, digital fulfilment, and operations. ○ Developed an available-to-promise (ATP) simulator from scratch—spanning algorithm design, data modelling, interface design, and deployment automation—and integrated it with the guest order allocation (GOA) simulator to evaluate end-to-end delivery speed, node selection, and shipping-cost trade-offs. ○ Validated the ATP simulator through large-scale backtesting against production behaviour, improving accuracy from about 40% to 85% and establishing it as a trusted testbed for promise and fulfilment changes. ○ Modernised data ingestion with Kafka-based pipelines and scalable PySpark/Hive processing; added input validation, scenario templates, and automated reporting; and improved runtime by 40% through exhaustive profiling and optimisation. ○ Delivered simulation studies informing decisions affecting roughly 3% of Target’s digital orders (\$600M+), including a FedEx SmartPost evaluation that identified up to \$5M in potential shipping-cost savings for an estimated 0.03-day increase in delivery time. ○ Owned the roadmap across five major and 12 minor releases in 2025–2026, including automated accuracy measurement and reporting; presented the platform’s evolution and scaling roadmap in “Scaling Nemo: Experiences and Challenges” at an organisation-wide Simulations Conference.
2021 – 2024	Past Projects, Target, WNS, IIMB, and Personal. <i>A list of selected past projects: selected_past_projects_CV.pdf</i>

Tools and Skills

Expert	Programming & Data Engineering , Python, R, PySpark, SQL, Shiny/Streamlit, C/C++.
Advanced	Analytics & Modelling , Regression and classification, experiment design, A/B testing, NLP, feature engineering, MLOps fundamentals, H2O, TensorFlow/Keras.
Advanced	Generative AI & Agentic AI , Multi-agent orchestration, tool calling, Model Context Protocol (MCP), LLM application design, prompt, skill, and agent engineering, AI governance and observability, specification-driven development, OpenCode, GitHub Copilot.
Expert	Design Principles , Data structures and algorithms, time and space complexity, vectorisation, functional programming, test-driven development, modular development, secure coding practices.
Advanced	Engineering & Platforms , APIs and services (Plumber/FastAPI), basic Kotlin/Java, debugging, web scraping (Selenium), Linux, code review, Git/GitHub, relational databases, Hadoop/Hive, Kafka, AWS, CI/CD.
Advanced	Visualisation & Storytelling , ggplot2, Plotly, R Markdown, dashboarding, LaTeX, Microsoft Office, Confluence/Jira.
Expert	Domain Knowledge , Last-mile operations research (available-to-promise and order allocation), digital fulfilment and shipping-cost optimisation, finance, accounting, and statistics.
Advanced	Product & Strategy , Problem solving, product thinking, scoping ambiguous problems, solution design and impact measurement, strategic thinking.
Advanced	Leadership , Mentoring and coaching, project evaluation and planning, team management, stakeholder management, product leadership, hiring and interviewing.

Education

- 2016 – **Doctor of Philosophy**, *Finance and Accounting*, Indian Institute of Management,
2021 Bangalore.
CGPA: 3.64 / 4.00
- 2010 – **Master of Engineering**, *Electrical Communication Engineering*, Indian Institute of
2012 Science, Bangalore.
CGPA: 6.4 / 8.0; Project Grade: A
- 2006 – **Bachelor of Technology**, *Electronics and Communication Engineering*, Bundelkhand
2010 Institute of Engineering and Technology, Jhansi (U.P.).
Grade: 73.6%
- 2004 – **Schooling (12th Standard)**, *Science*, C.B.S.E..
2006 Grade: 88.2%

Awards and Honours

- 2025 **Drive**, *Target Corporation*.
- 2022 **Tech Genius**, *Transforming the Organization*, **Victory Fleet**, *WNS Global Services*.
- 2020 **Mirae Asset Scholarship (PhD Year 5)**, *Indian Institute of Management*.
- 2017–2018 **Director's Merit List (PhD Years 1 and 2)**, *Indian Institute of Management*.
- 2016 **96.5th percentile**, *Common Aptitude Test (CAT)*.
- 2010 **All India Rank 20 of 105,000**, *Graduate Aptitude Test in Engineering (GATE)*.
- 2006–2007 **Merit Scholarships**, *Intermediate Examination and Engineering*.